



VOCERA ANALYTICS AI DIAGNOSTICS AGENT

An Agentic AI initiative

Abstract

The Vocera Analytics AI Diagnostics Agent is an AI-powered troubleshooting system designed to significantly reduce Mean Time To Resolution (MTTR)

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Vocera Analytics AI Diagnostics Agent

Detailed Technical Dossier

1. Executive Summary

The **Vocera Analytics AI Diagnostics Agent** is an AI-powered troubleshooting system designed to significantly reduce **Mean Time To Resolution (MTTR)**.

It combines:

- Real-time configuration monitoring
- Intelligent log ingestion
- AI-driven root cause analysis

This enables engineers to diagnose and resolve complex issues across **MTLS, database systems, and distributed components** within minutes instead of hours.

2. Problem Statement

Vocera Analytics operates in a **highly distributed and security-sensitive environment**, leading to:

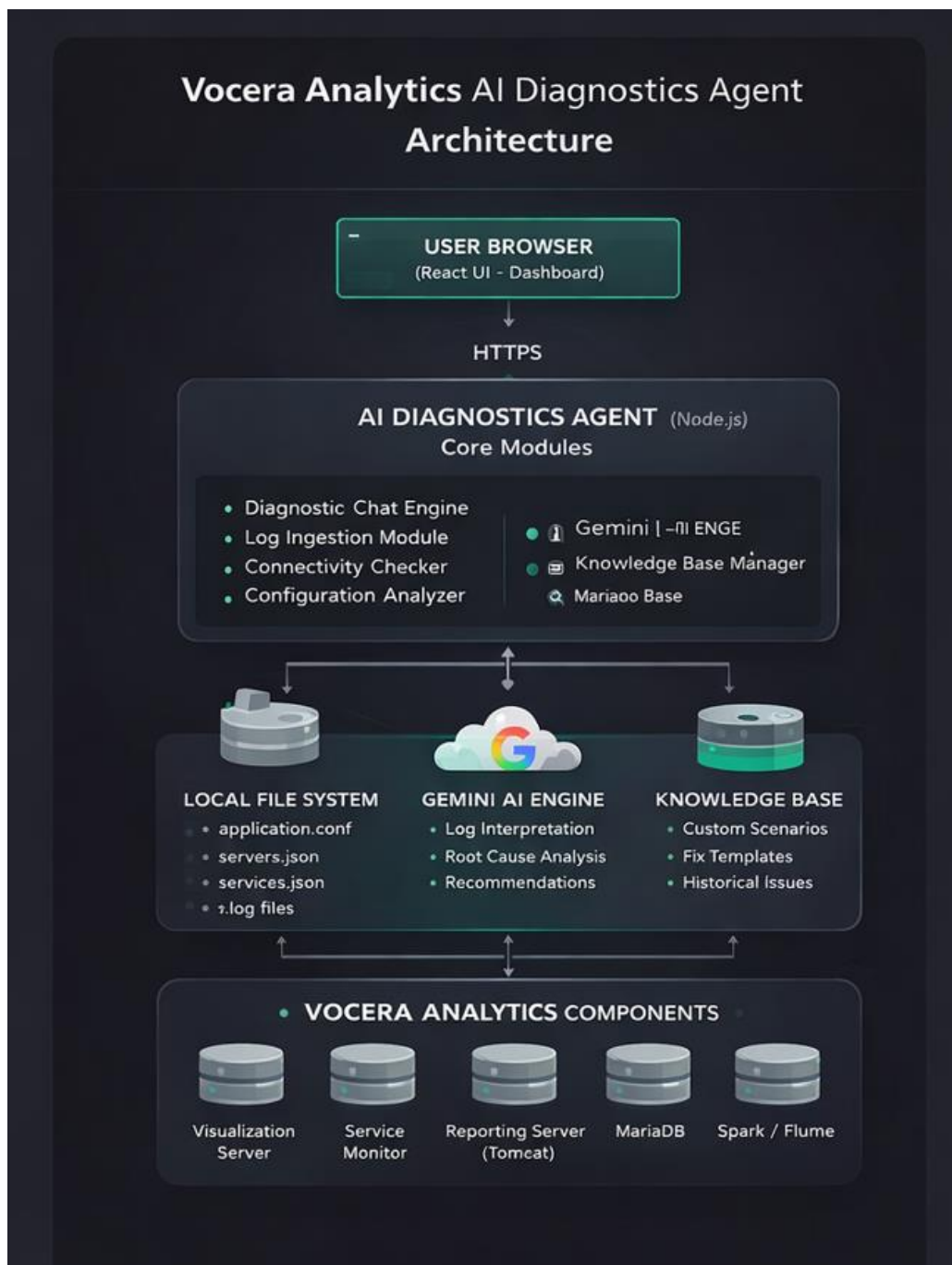
- **Manual Log Analysis**
Engineers sift through large logs like Catalina.out, VoceraVA.log, etc.
- **Configuration Blind Spots**
Issues like expired CRLs or misconfigured MTLS go unnoticed
- **Knowledge Silos**
Critical fixes are known only to senior engineers

👉 Result:

High MTTR + Operational inefficiency

3. Architecture Overview

The AI Diagnostics Agent follows a **lightweight, sidecar architecture**:



Components:

- **Frontend:** React-based UI (Mission Control Dashboard)
- **Backend:** Node.js (File system interaction)
- **AI Engine:** Google Gemini (log interpretation)
- **Data Sources:**
 - application.conf
 - servers.json
 - Log files (*.log)
 - CRL files (VoceraVA.crl)

Key Design Principle:

- ✓ Read-only
 - ✓ Zero impact on production
 - ✓ Deployable in minutes
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4. Workflow (Diagnostic Loop)



The system follows a **closed-loop intelligent workflow**:

1. **Ingest**

- Logs (upload/paste)
- Config files

2. **Analyze**

- AI correlates logs with system configuration

3. **Diagnose**

- Root cause identified (e.g., timeout, MTLS failure)

4. **Resolve**

- Step-by-step fix provided
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5. Application Screens

5.1 Diagnostic Chat

Purpose:

Primary interface for AI-driven troubleshooting

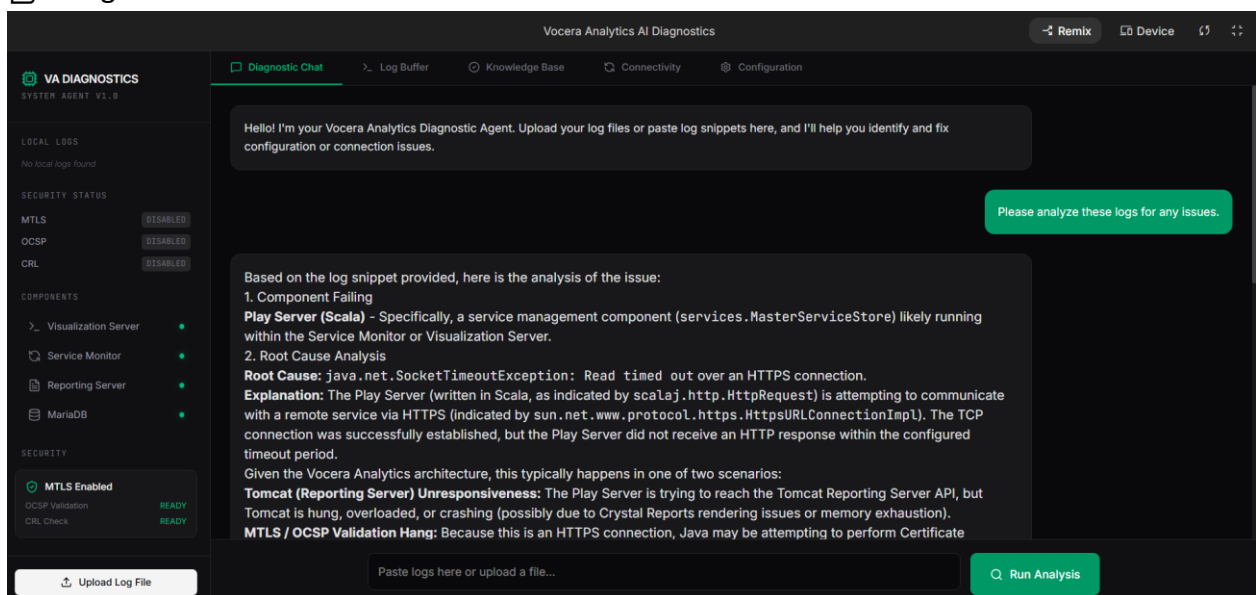
Key Capabilities:

- Accepts log inputs
- Performs contextual AI analysis
- Provides:
 - Root cause
 - Explanation
 - Resolution steps

Example Insight:

- Detects SocketTimeoutException
- Suggests:
 - Check Reporting Server
 - Validate MTLS/OCSP

Diagnostic Chat screen -



5.2 Log Buffer

Purpose:

Raw log ingestion interface

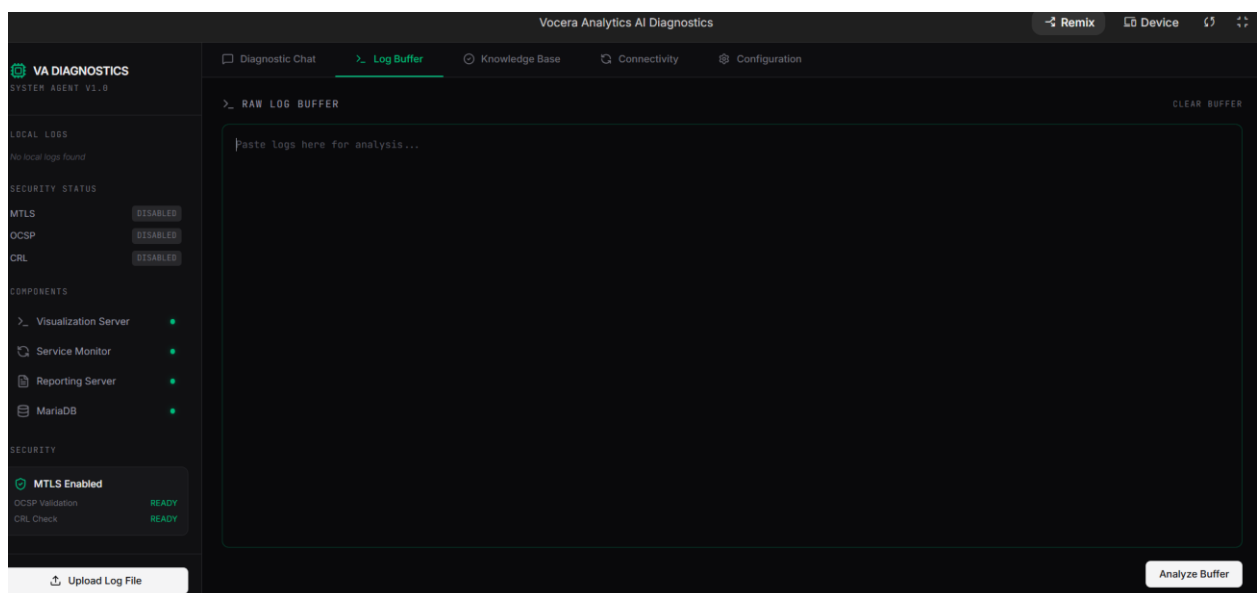
Key Features:

- Paste large log files
- Upload logs directly
- Trigger analysis via “Analyze Buffer”

Role in System:

👉 Entry point for **Ingest** phase

👉 **Log Buffer** screen –



5.3 Knowledge Base

Purpose:

Central repository of troubleshooting scenarios

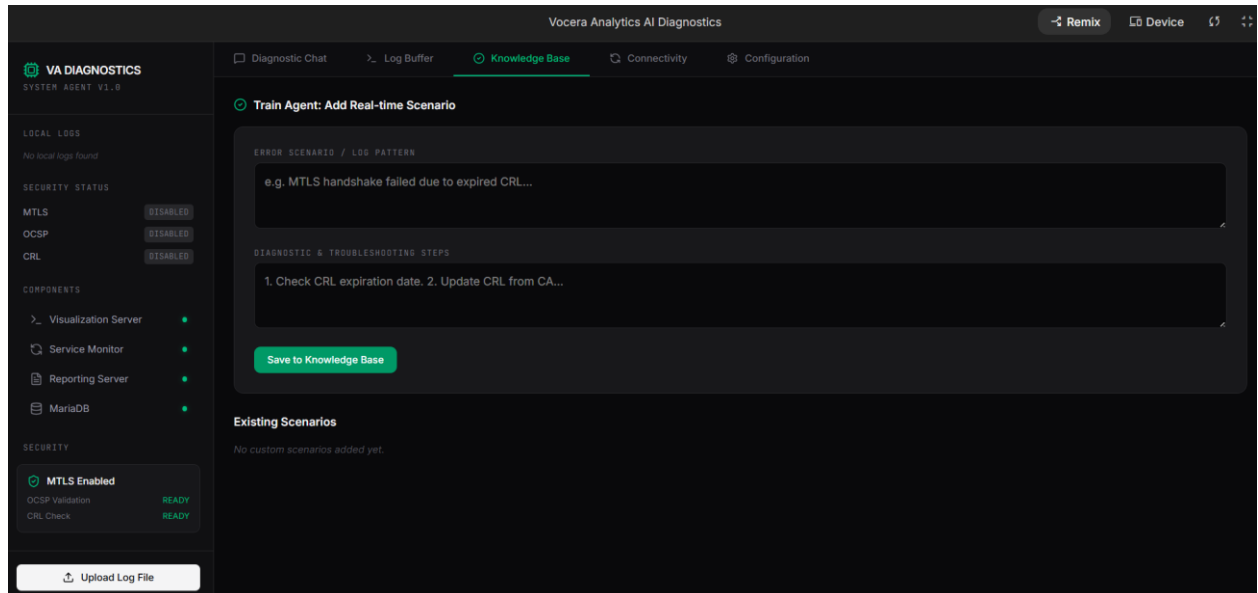
Key Features:

- Store solutions
- Reuse past fixes
- Train the AI agent

Impact:

👉 Eliminates dependency on tribal knowledge

👉 **Knowledge Base screen**

**5.4 Connectivity****Purpose:**

Network diagnostics module

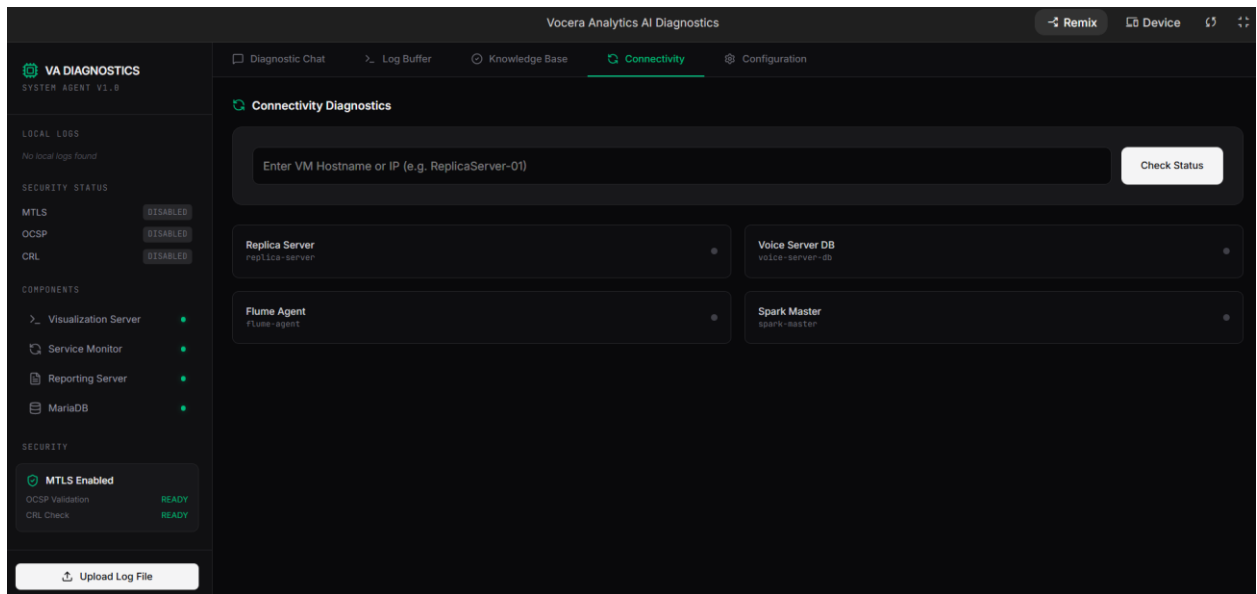
Key Features:

- VM-to-VM connectivity checks
- One-click validation
- Raw output visibility

Use Case:

👉 Debug replica/voice server communication issues

📁 Connectivity Tab screen of VA Diagnostics Agent



5.5 Configuration

Purpose:

Real-time system and security monitoring

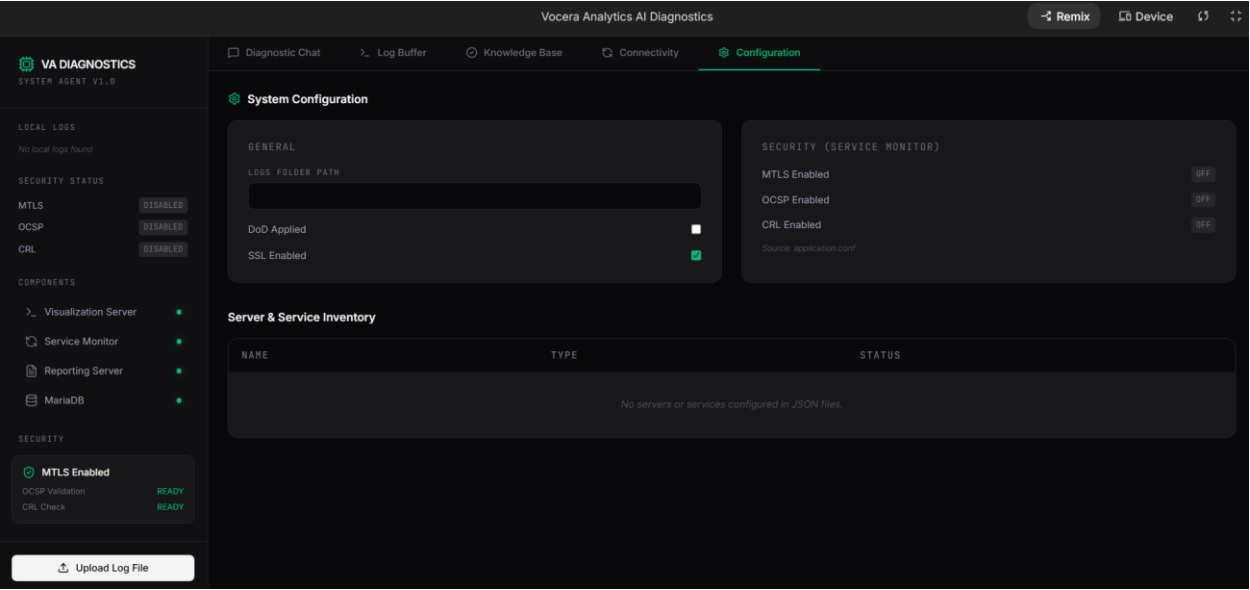
Key Features:

- MTLS status
- OCSP validation
- CRL check
- Component health

Critical Value:

📁 Detects issues **before failures occur**

📁 Configuration screen



6. Business Value & ROI

Key Benefits:

-  **80% faster diagnostics**
-  **Skill leveling (Tier 1 → Tier 3)**
-  **Proactive security monitoring**
-  **Improved system uptime**

Before vs After:

Aspect	Before	After
Log Analysis	Manual	AI-driven
Root Cause	Hours	Minutes
Dependency	Senior Engineers	All Engineers

7. Integration & Safety

-  Read-only access
-  Sidecar deployment
-  No production impact
-  Minimal CPU/RAM usage
-  Air-gap compatible

8. Conclusion

The **AI Diagnostics Agent** transforms Vocera Analytics troubleshooting from:

- ✗ Reactive →  Intelligent
- ✗ Manual →  Automated
- ✗ Slow →  Real-time

 Enabling faster, smarter, and more reliable operations.